

ELITE MATHEMATICS

Pearson Edexcel International A Level

Mechanics 1

Adaptive Classified

Chapter 10 • Moments

Questions

Student book • questions with writing space

- 42 pages
- Past papers up to June 2026

MECHANICS 1 | TOPIC 10

Moments

Your classified route

19

QUESTIONS

147

MARKS

3

FAMILIES

9

VARIANTS

DR ESLAM AHMED

CLASSIFIED BY IDEA

Read the diagram, choose the model, then use the working space to complete the mechanics.

TOPIC 10 | FAMILY 01

Beam/rod equilibrium and support forces

WHAT THIS FAMILY ASKS

Combine moments and vertical equilibrium to find reactions, tensions or attached loads.

VARIANT ROUTE**V01** Two supports or strings**V02** Given reaction/tension ratio**V03** Attached particle or package**V04** Support or rope capacity limits an attached load**V05** Admissible range from non-negative reactions**10 classified questions**

83 total marks

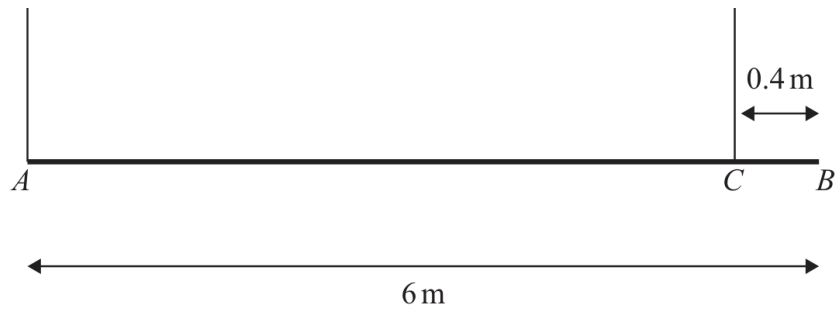


Figure 2 shows a uniform rod AB of length 6 m suspended by two light vertical ropes.

The first rope is attached to the rod at A .

The second rope is attached to the rod at the point C , where $CB = 0.4$ m.

The rod has mass 30 kg.

A particle of mass 20 kg is now attached to the rod at B .

The rod is in equilibrium in a horizontal position.

(a) Find

- (i) the tension in the rope attached to the rod at C ,
- (ii) the tension in the rope attached to the rod at A .

(6)

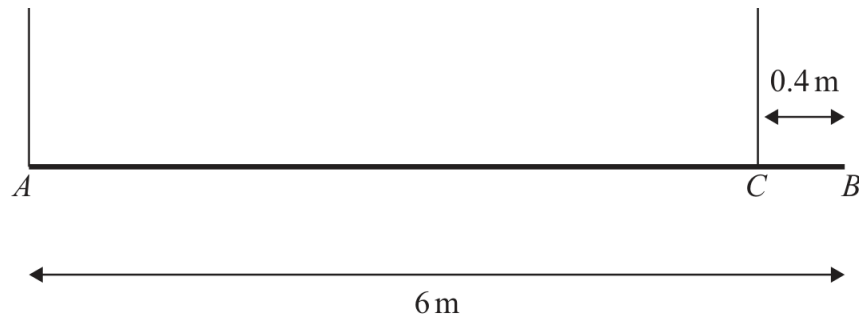


Figure 2 shows a uniform rod AB of length 6 m suspended by two light vertical ropes.

The first rope is attached to the rod at A .

The second rope is attached to the rod at the point C , where $CB = 0.4$ m.

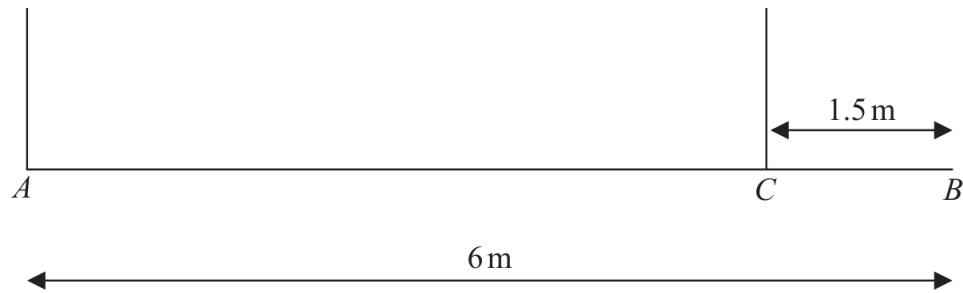
The rod has mass 30 kg.

The particle of mass 20 kg at B is removed and replaced by a particle of mass M kg.

The rod remains in equilibrium in a horizontal position.

(b) Find the exact maximum value of M .

(3)

**Figure 1**

A non-uniform beam AB has length 6 m and weight W newtons. The beam is supported in equilibrium in a horizontal position by two vertical ropes, one attached to the beam at A and the other attached to the beam at C , where $CB = 1.5$ m, as shown in Figure 1.

The centre of mass of the beam is 2.625 m from A .

The ropes are modelled as light strings. The beam is modelled as a non-uniform rod.

Given that the tension in the rope attached at C is 20 N greater than the tension in the rope attached at A ,

(a) find the value of W .

(6)

A non-uniform beam AB has length 6 m and weight W newtons. The beam is supported in equilibrium in a horizontal position by two vertical ropes, one attached to the beam at A and the other attached to the beam at C , where $CB = 1.5$ m, as shown in Figure 1.

The centre of mass of the beam is 2.625 m from A .

The ropes are modelled as light strings. The beam is modelled as a non-uniform rod.

(b) State how you have used the fact that the beam is modelled as a rod.

(1)

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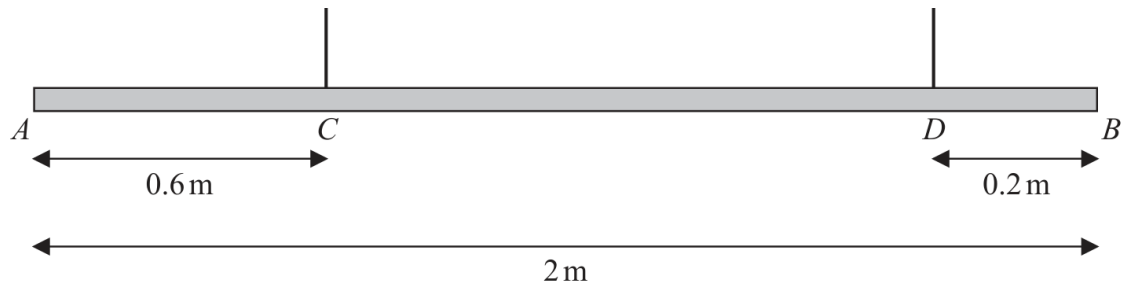
**Figure 1**

Figure 1 shows a beam AB , of mass m kg and length 2 m, suspended by two light vertical ropes.

The ropes are attached to the points C and D on the beam, where $AC = 0.6$ m and $DB = 0.2$ m

The beam is in equilibrium in a horizontal position.

A particle of mass pm kg is attached to the beam at A and the beam remains in equilibrium in a horizontal position.

The beam is modelled as a uniform rod.

- (a) Given that the tension in the rope attached at C is four times the tension in the rope attached at D , use the model to find the exact value of p .

(7)

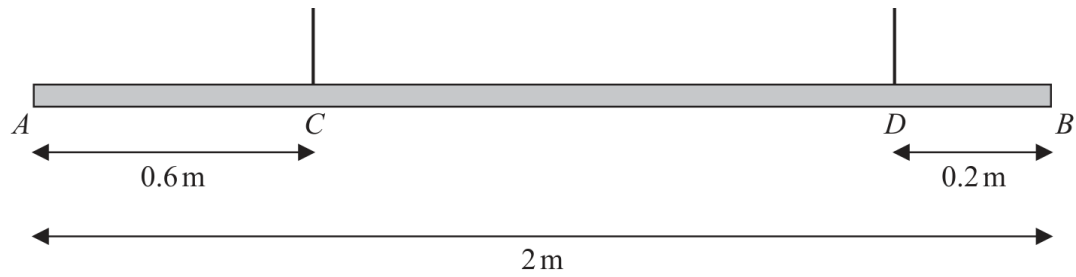
**Figure 1**

Figure 1 shows a beam AB , of mass m kg and length 2 m, suspended by two light vertical ropes.

The ropes are attached to the points C and D on the beam, where $AC = 0.6$ m and $DB = 0.2$ m

The beam is in equilibrium in a horizontal position.

The beam is modelled as a uniform rod.

The particle of mass pm kg at A is removed and replaced by a particle of mass qm kg at A .

The beam remains in equilibrium in a horizontal position but is now on the point of tilting.

(b) Using the model, find the exact value of q

(4)

Figure 1 shows a beam AB , of mass m kg and length 2 m, suspended by two light vertical ropes.

The ropes are attached to the points C and D on the beam, where $AC = 0.6$ m and $DB = 0.2$ m

The beam is in equilibrium in a horizontal position.

The beam is modelled as a uniform rod.

(c) State how you have used the modelling assumption that the beam is uniform.

(1)

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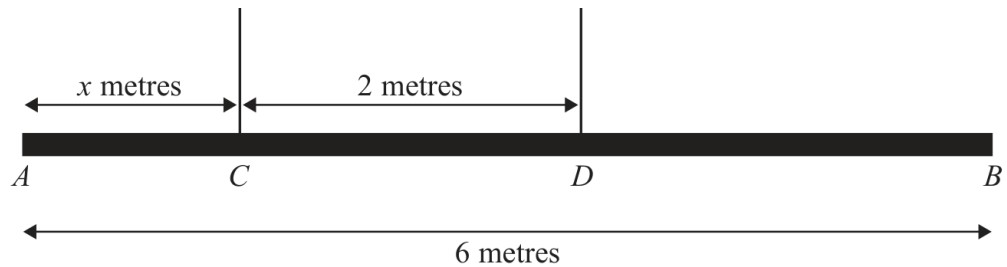
**Figure 1**

Figure 1 shows a beam AB with weight 24 N and length 6 m .

The beam is suspended by two light vertical ropes. The ropes are attached to the points C and D on the beam where $AC = x$ metres and $CD = 2\text{ m}$.

The tension in the rope attached to the beam at C is double the tension in the rope attached to the beam at D .

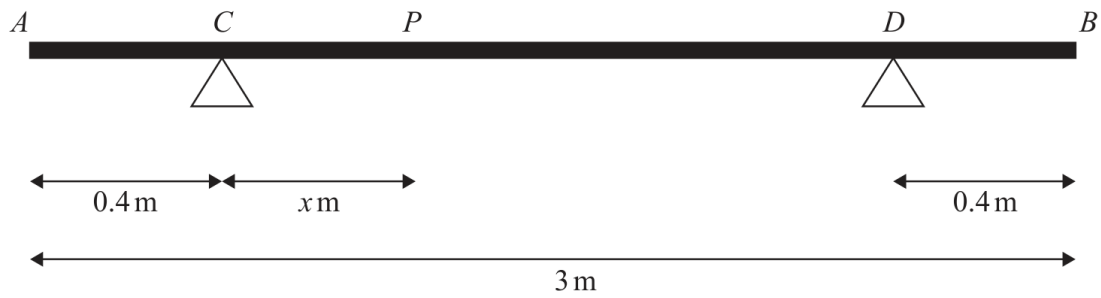
The beam is modelled as a uniform rod, resting horizontally in equilibrium.

Find

- (i) the tension in the rope attached to the beam at D .
- (ii) the value of x .

(5)

5.



A beam AB has mass 30 kg and length 3 m.

The beam rests on supports at C and D where $AC = 0.4$ m and $DB = 0.4$ m, as shown in Figure 4.

A person of mass 55 kg stands on the beam between C and D .

The person is modelled as a particle at the point P , where $CP = x$ metres and $0 < x < 2.2$

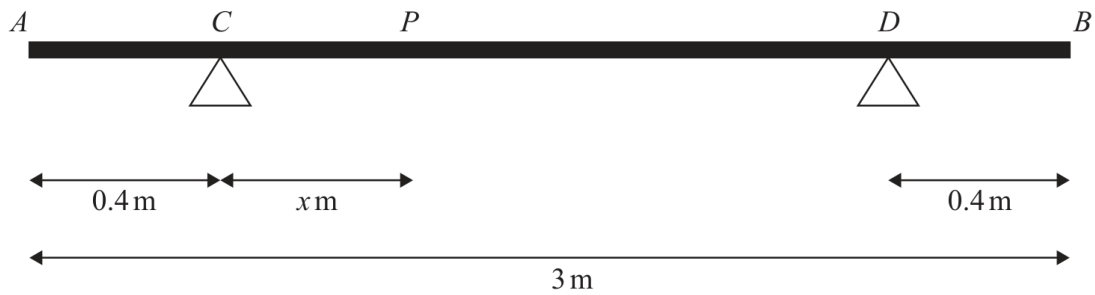
The beam is modelled as a uniform rod resting in equilibrium in a horizontal position.

Using the model,

(a) show that the magnitude of the reaction at C is $(686 - 245x)$ N.

(3)

5.



A beam AB has mass 30 kg and length 3 m.

The beam rests on supports at C and D where $AC = 0.4$ m and $DB = 0.4$ m, as shown in Figure 4.

A person of mass 55 kg stands on the beam between C and D .

The person is modelled as a particle at the point P , where $CP = x$ metres and $0 < x < 2.2$

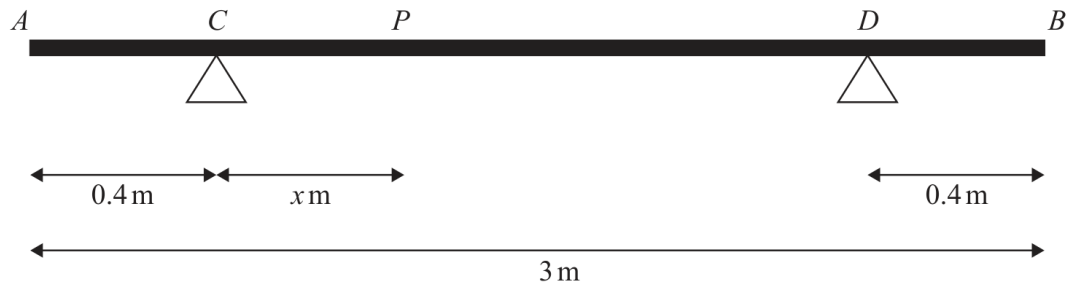
The magnitude of the reaction at C is **four** times the magnitude of the reaction at D .

Using the model,

(b) find the value of x

(4)

5.



A beam AB has mass 30 kg and length 3 m .

The beam rests on supports at C and D where $AC = 0.4\text{ m}$ and $DB = 0.4\text{ m}$, as shown in Figure 4.

The beam is modelled as a uniform rod resting in equilibrium in a horizontal position.

The person steps off the beam and places a package of mass $M\text{ kg}$ at A .

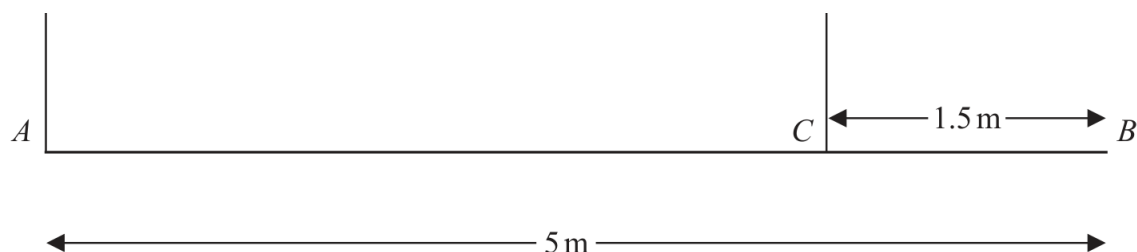
The package is modelled as a particle at the point A .

The beam is now on the point of tilting about C .

Using the model,

(c) find the value of M

(3)

**Figure 2**

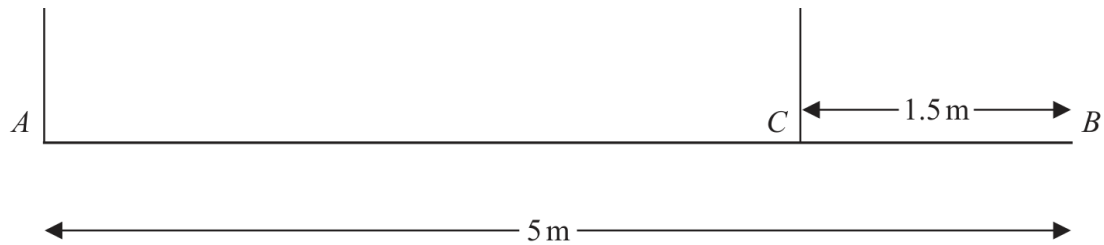
A non-uniform beam, AB , has length 5 m and mass 12 kg. The beam is suspended in a horizontal position by two vertical ropes. One rope is attached to the beam at A . The other rope is attached to the beam at C , where $CB = 1.5$ m, as shown in Figure 2. The distance of the centre of mass of the beam from A is 1.75 m. The beam is modelled as a non-uniform rod and the ropes are modelled as light inextensible strings.

A particle of mass M kg is now placed on the beam at B and the beam remains in equilibrium in a horizontal position.

- (a) Find the largest possible value of M .

(3)

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**Figure 2**

A non-uniform beam, AB , has length 5 m and mass 12 kg. The beam is suspended in a horizontal position by two vertical ropes. One rope is attached to the beam at A . The other rope is attached to the beam at C , where $CB = 1.5$ m, as shown in Figure 2. The distance of the centre of mass of the beam from A is 1.75 m. The beam is modelled as a non-uniform rod and the ropes are modelled as light inextensible strings.

The particle at B is now removed and a particle of mass 15 kg is now placed on the beam at the point D , where $AD = x$ metres. The beam remains in equilibrium in a horizontal position.

Given that the tension in the rope attached to the beam at C is now twice the tension in the rope attached to the beam at A ,

(b) find the value of x .

(5)

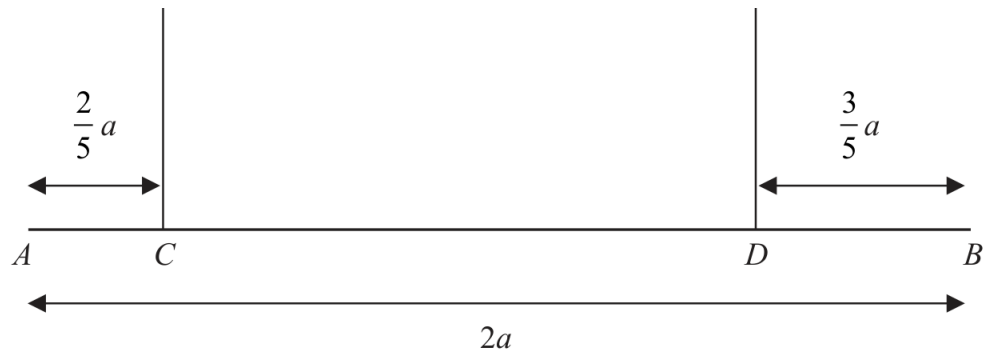


Figure 1

A uniform rod AB has length $2a$ and mass M . The rod is held in equilibrium in a horizontal position by two vertical light strings which are attached to the rod at C and D ,

where $AC = \frac{2}{5}a$ and $DB = \frac{3}{5}a$, as shown in Figure 1.

A particle P is placed on the rod at B .

The rod remains horizontal and in equilibrium.

(a) Find, in terms of M , the largest possible mass of the particle P

(3)

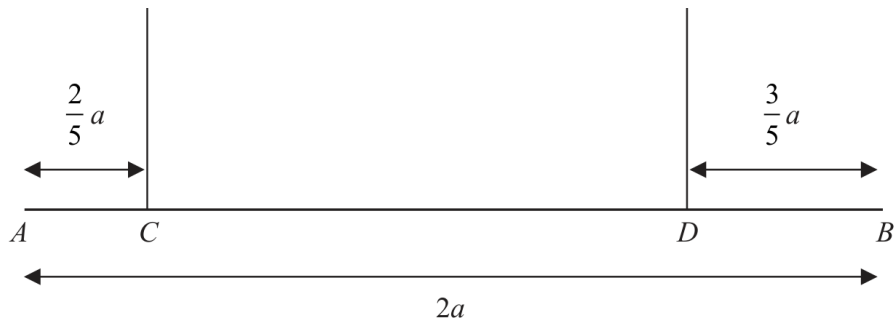


Figure 1

A uniform rod AB has length $2a$ and mass M . The rod is held in equilibrium in a horizontal position by two vertical light strings which are attached to the rod at C and D ,

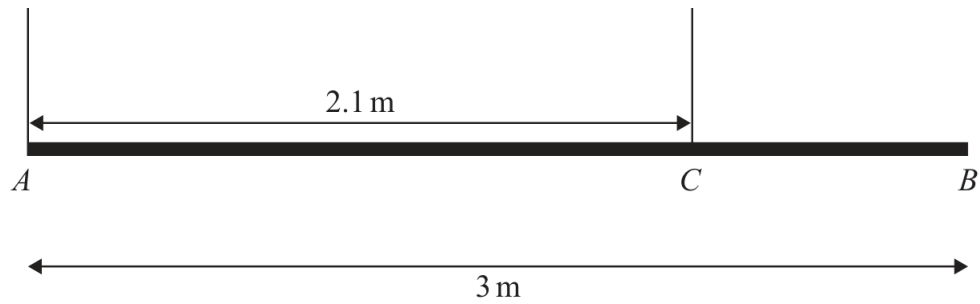
where $AC = \frac{2}{5}a$ and $DB = \frac{3}{5}a$, as shown in Figure 1.

A particle P is placed on the rod at B .

The rod remains horizontal and in equilibrium.

Given that the mass of P is $\frac{1}{2}M$

(b) find, in terms of M and g , the tension in the string that is attached to the rod at C . (3)

**Figure 2**

A uniform rod AB has length 3 m and weight W newtons.

The rod is suspended by two light vertical ropes.

One rope is attached to the rod at A and the other rope is attached to the rod at C , where $AC = 2.1$ m.

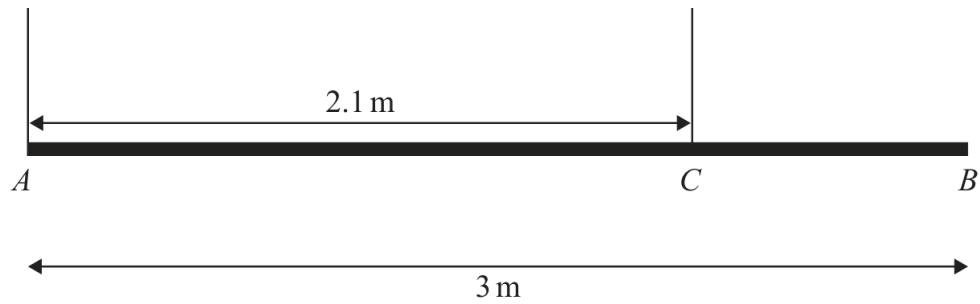
The rod is in equilibrium in a horizontal position, as shown in Figure 2.

The tension in the rope at C is 350 N.

(a) Show that $W = 490$

(3)

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**Figure 2**

A uniform rod AB has length 3 m and weight W newtons.

The rod is suspended by two light vertical ropes.

One rope is attached to the rod at A and the other rope is attached to the rod at C , where $AC = 2.1$ m.

The rod is in equilibrium in a horizontal position, as shown in Figure 2.

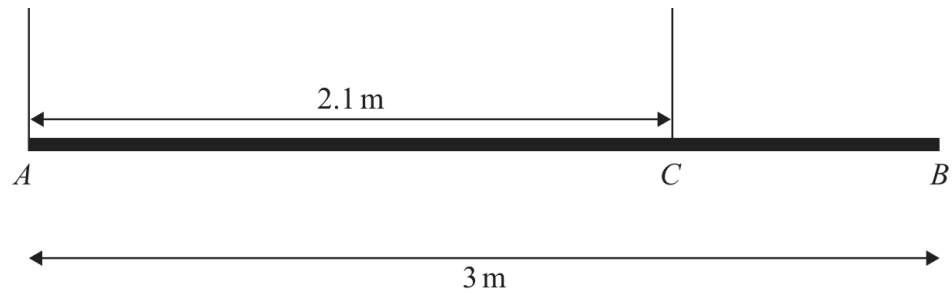
A particle P of weight 210 N is attached to the rod at a distance d metres from A .

The tension in the rope at C is now 600 N.

The rod remains in equilibrium in a horizontal position.

(b) Find the value of d .

(3)

**Figure 2**

A uniform rod AB has length 3 m and weight W newtons.
 The rod is suspended by two light vertical ropes.
 One rope is attached to the rod at A and the other rope is attached to the rod at C ,
 where $AC = 2.1\text{ m}$.

The rod is in equilibrium in a horizontal position, as shown in Figure 2.

Particle P is removed from the rod.

A particle Q of weight X newtons is now attached at B .

The rod remains in equilibrium in a horizontal position and is now on the point of tilting.

(c) Find the value of X .

(4)

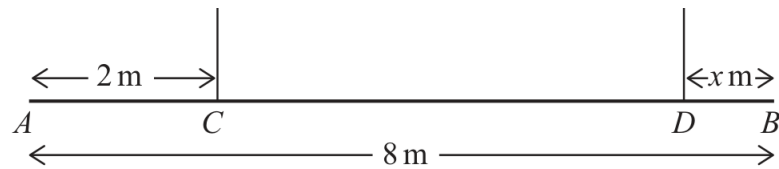
**Figure 1**

Figure 1 shows a sketch of a beam AB , with weight 240 N and length 8 m .

The beam is held in equilibrium in a horizontal position by two vertical ropes. The ropes are attached to the beam at the points C and D , where $AC = 2\text{ m}$ and $DB = x$ metres.

The beam is modelled as a uniform rod and the ropes are modelled as light inextensible strings.

The tension in the rope at D is 90 N .

(a) Show that $x = \frac{2}{3}$

(3)

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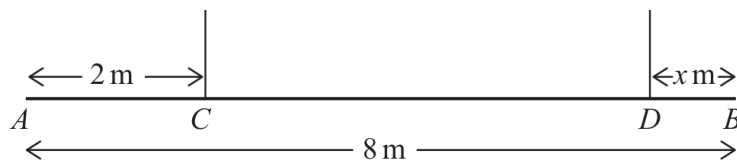
**Figure 1**

Figure 1 shows a sketch of a beam AB , with weight 240 N and length 8 m .

The beam is held in equilibrium in a horizontal position by two vertical ropes. The ropes are attached to the beam at the points C and D , where $AC = 2\text{ m}$ and $DB = x$ metres.

The beam is modelled as a uniform rod and the ropes are modelled as light inextensible strings.

The rope at C will break if its tension exceeds 183 N . The rope at D cannot break. A package of weight W newtons is now attached to the beam at A . The beam remains horizontal and in equilibrium.

The package is modelled as a particle.

It is given that the rope at C does not break.

(b) Find the greatest possible value of W .

(4)

5.

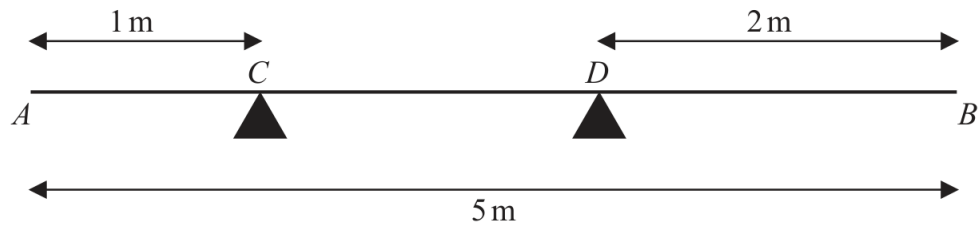


Figure 2

A uniform rod AB has length 5 m and mass 5 kg. The rod rests in equilibrium in a horizontal position on two supports C and D , where $AC = 1$ m and $DB = 2$ m, as shown in Figure 2.

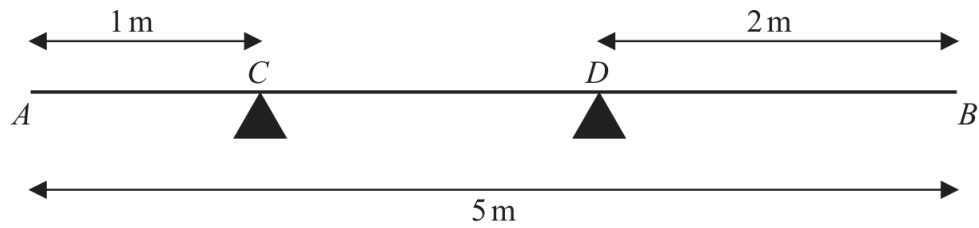
A particle of mass 10 kg is placed on the rod at A and a particle of mass M kg is placed on the rod at B . The rod remains horizontal and in equilibrium.

(a) Find, in terms of M , the magnitude of the reaction on the rod at C .

(3)

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5.

**Figure 2**

A uniform rod AB has length 5 m and mass 5 kg. The rod rests in equilibrium in a horizontal position on two supports C and D , where $AC = 1$ m and $DB = 2$ m, as shown in Figure 2.

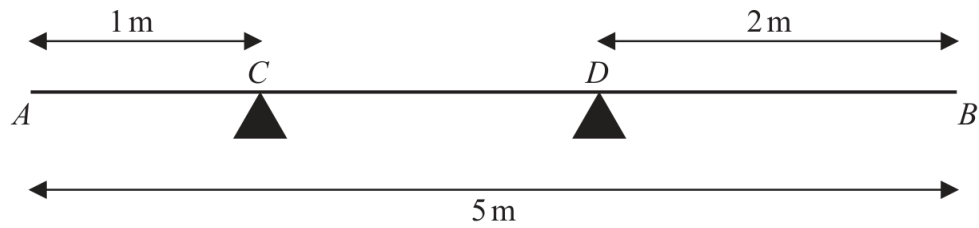
A particle of mass 10 kg is placed on the rod at A and a particle of mass M kg is placed on the rod at B . The rod remains horizontal and in equilibrium.

(b) Find, in terms of M , the magnitude of the reaction on the rod at D .

(3)

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5.

**Figure 2**

A uniform rod AB has length 5 m and mass 5 kg. The rod rests in equilibrium in a horizontal position on two supports C and D , where $AC = 1$ m and $DB = 2$ m, as shown in Figure 2.

A particle of mass 10 kg is placed on the rod at A and a particle of mass M kg is placed on the rod at B . The rod remains horizontal and in equilibrium.

(c) Hence, or otherwise, find the range of possible values of M .

(3)

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TOPIC 10 | FAMILY 02

Locate the centre of mass of a rod

WHAT THIS FAMILY ASKS

Infer or use the position of a rod's weight, especially for non-uniform rods.

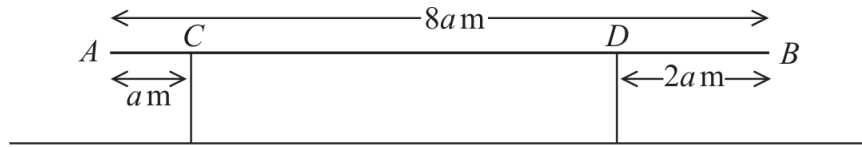
VARIANT ROUTE

V01 Infer centre from reaction data

3 classified questions

25 total marks

7.

**Figure 2**

A non-uniform beam AB , of mass 60 kg and length $8a$ metres, rests in equilibrium in a horizontal position on two vertical supports. One support is at C , where $AC = a$ metres and the other support is at D , where $DB = 2a$ metres, as shown in Figure 2.

The magnitude of the normal reaction between the beam and the support at D is three times the magnitude of the normal reaction between the beam and the support at C .

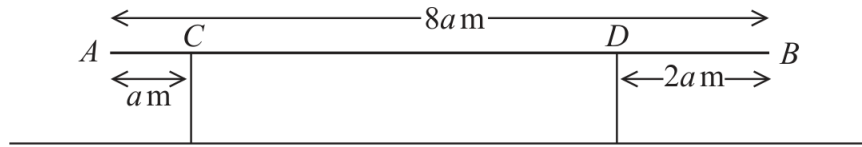
By modelling the beam as a non-uniform rod whose centre of mass is at a distance x metres from A ,

- (a) find an expression for x in terms of a .

(5)

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7.

**Figure 2**

A non-uniform beam AB , of mass 60 kg and length $8a$ metres, rests in equilibrium in a horizontal position on two vertical supports. One support is at C , where $AC = a$ metres and the other support is at D , where $DB = 2a$ metres, as shown in Figure 2.

A box of mass $M\text{ kg}$ is placed on the beam at E , where $AE = 2a$ metres.

The beam remains in equilibrium in a horizontal position.

The magnitude of the normal reaction between the beam and the support at C is now equal to the magnitude of the normal reaction between the beam and the support at D .

By modelling the box as a particle,

(b) find the value of M .

(5)

1.

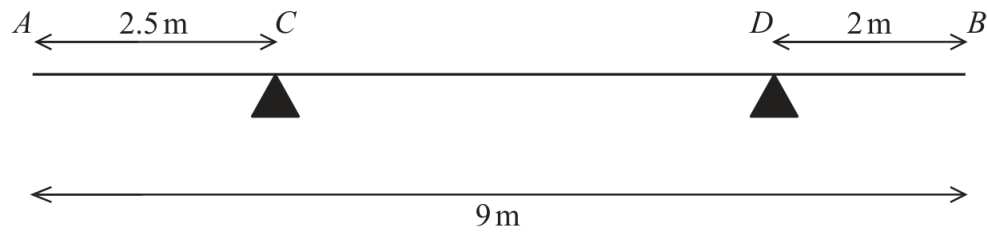


Figure 1

A non-uniform rod AB has length 9 m and mass M kg.

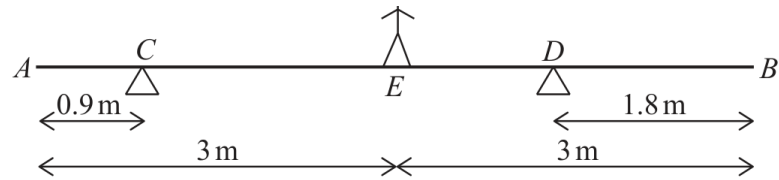
The rod rests in equilibrium in a horizontal position on two supports, one at C where $AC = 2.5$ m and the other at D where $DB = 2$ m, as shown in Figure 1.

The magnitude of the force acting on the rod at D is twice the magnitude of the force acting on the rod at C .

The centre of mass of the rod is d metres from A .

Find the value of d .

(6)

**Figure 1**

A non-uniform beam AB has length 6 m and mass 50 kg. The beam rests horizontally on two supports at C and D , where $AC = 0.9$ m and $DB = 1.8$ m.

A child of mass 25 kg stands on the beam at E , where $AE = EB = 3$ m, as shown in Figure 1.

The magnitude of the normal reaction between the beam and the support at C is R_C newtons.

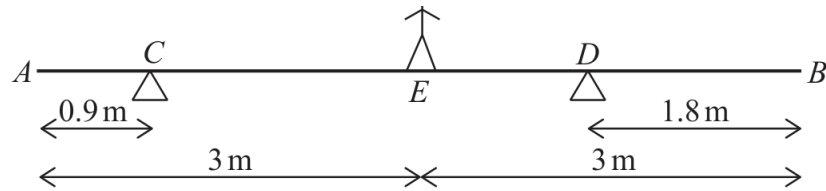
The magnitude of the normal reaction between the beam and the support at D is R_D newtons.

The centre of mass of the beam is between C and D and is a distance x metres from D .

Given that $2R_D = 3R_C$

(a) show that $x = 1.38$

(6)

**Figure 1**

A non-uniform beam AB has length 6 m and mass 50 kg. The beam rests horizontally on two supports at C and D , where $AC = 0.9$ m and $DB = 1.8$ m.

A child of mass 25 kg stands on the beam at E , where $AE = EB = 3$ m, as shown in Figure 1.

The child remains at E and a block of mass M kg is placed on the beam at B .

The block is modelled as a particle.

Given that the beam is on the point of tilting,

(b) find the value of M .

(3)

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TOPIC 10 | FAMILY 03

Impending tilt and loss of support

WHAT THIS FAMILY ASKS

Use the boundary condition that one support force becomes zero and moments balance about the remaining pivot.

VARIANT ROUTE

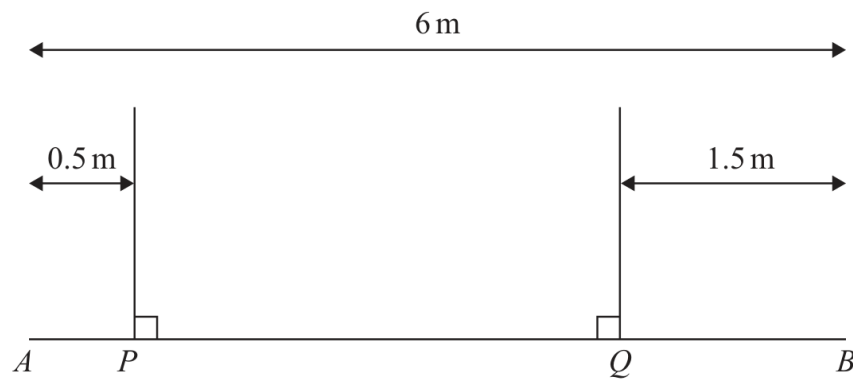
V01 One support becomes slack

V02 Overhang balanced by a counterweight

V03 Two alternative tipping configurations

6 classified questions

39 total marks

**Figure 1**

A non-uniform rod AB has mass 20 kg and length 6 m . The rod is held in equilibrium in a horizontal position by two light vertical strings. One string is attached to the rod at P , where $AP = 0.5\text{ m}$ and the other is attached to the rod at Q , where $QB = 1.5\text{ m}$, as shown in Figure 1.

The centre of mass of the rod is 4 m from A .

- (a) Find the tension in the string attached to the rod at Q .

(3)

**Figure 1**

A non-uniform rod AB has mass 20 kg and length 6 m . The rod is held in equilibrium in a horizontal position by two light vertical strings. One string is attached to the rod at P , where $AP = 0.5\text{ m}$ and the other is attached to the rod at Q , where $QB = 1.5\text{ m}$, as shown in Figure 1.

The centre of mass of the rod is 4 m from A .

A particle is now attached to the rod at B .

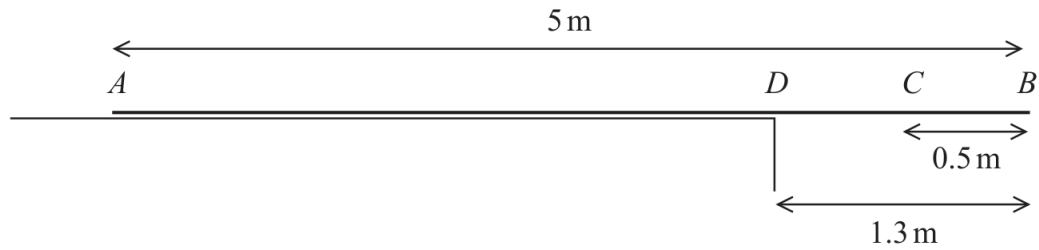
The rod remains in equilibrium in a horizontal position but is on the point of tilting.

The tension in the string attached to the rod at Q is now X newtons.

(b) Find the value of X .

(3)

3.

**Figure 2**

A beam $ADCB$ has length 5 m . The beam lies on a horizontal step with the end A on the step and the end B projecting over the edge of the step. The edge of the step is at the point D where $DB = 1.3\text{ m}$, as shown in Figure 2.

When a small boy of mass 30 kg stands on the beam at C , where $CB = 0.5\text{ m}$, the beam is on the point of tilting.

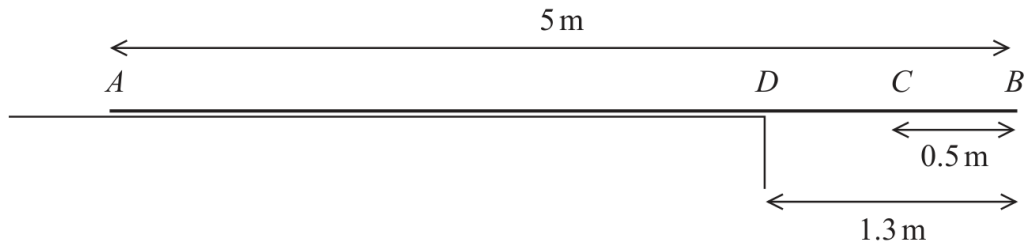
The boy is modelled as a particle and the beam is modelled as a uniform rod.

(a) Find the mass of the beam.

(3)

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3.

**Figure 2**

A beam $ADCB$ has length 5 m. The beam lies on a horizontal step with the end A on the step and the end B projecting over the edge of the step. The edge of the step is at the point D where $DB = 1.3$ m, as shown in Figure 2.

When a small boy of mass 30 kg stands on the beam at C , where $CB = 0.5$ m, the beam is on the point of tilting.

A block of mass X kg is now placed on the beam at A .

The block is modelled as a particle.

- (b) Find the smallest value of X that will enable the boy to stand on the beam at B without the beam tilting.

(3)

A block of mass X kg is now placed on the beam at A .

The block is modelled as a particle.

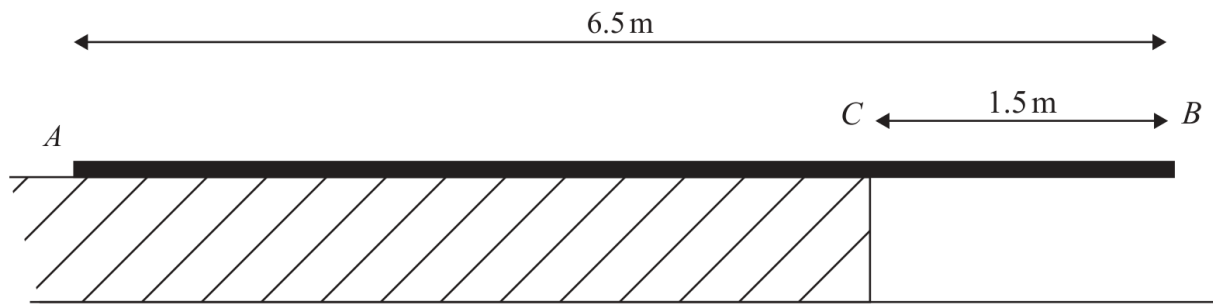
- (b) Find the smallest value of X that will enable the boy to stand on the beam at B without the beam tilting.

(3)

- (c) State how you have used the modelling assumption that the block is a particle in your calculations.

(1)

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**Figure 2**

A non-uniform rod AB has length 6.5 m and mass 1.2 kg. The centre of mass of the rod is 3 m from A . The rod rests on a horizontal step and overhangs the end of the step C by 1.5 m, as shown in Figure 2.

The rod is perpendicular to the edge of the step.

A particle of mass 4 kg is placed on the rod at B and another particle, whose mass is M kg, is placed on the rod at D , where $AD = 0.5$ m.

The rod remains in equilibrium in a horizontal position.

(a) Find the smallest possible value of M .

(3)

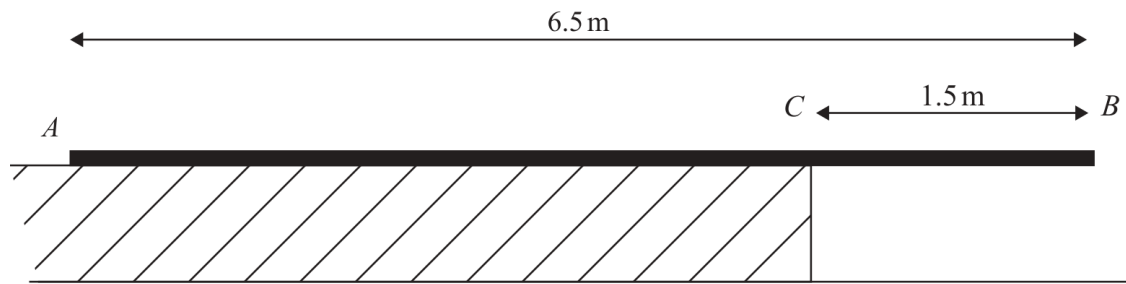


Figure 2

A non-uniform rod AB has length 6.5 m and mass 1.2 kg . The centre of mass of the rod is 3 m from A . The rod rests on a horizontal step and overhangs the end of the step C by 1.5 m , as shown in Figure 2.

The rod is perpendicular to the edge of the step.

The particle at B **and** the particle at D are now **removed**.

A new particle is placed on the rod at the point E , where $EB = 0.9\text{ m}$.

The rod remains in equilibrium in a horizontal position but is on the point of tilting about C .

(b) Find the magnitude of the force acting on the rod at C .

(3)

4.

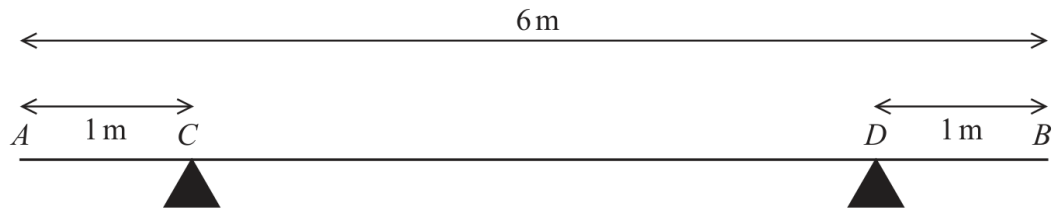


Figure 2

A metal girder AB has weight W newtons and length 6 m. The girder rests in a horizontal position on two supports C and D where $AC = DB = 1$ m, as shown in Figure 2.

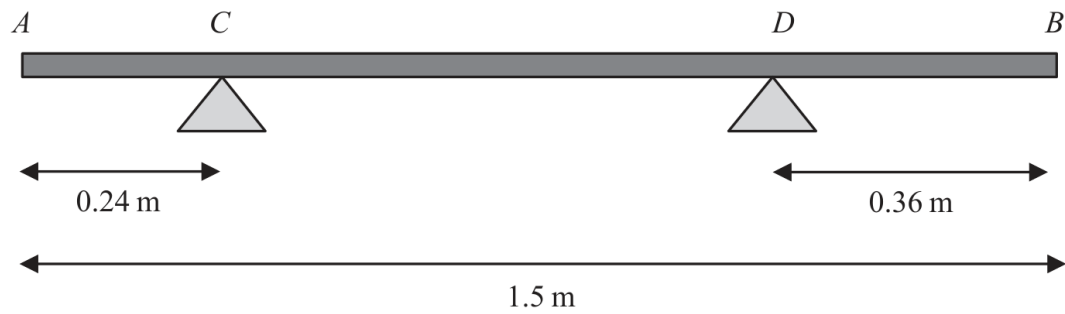
When a force of magnitude 900 N is applied vertically upwards to the girder at A , the girder is about to tilt about D .

When a force of magnitude 1500 N is applied vertically upwards to the girder at B , the girder is about to tilt about C .

The girder is modelled as a non-uniform rod whose centre of mass is a distance x metres from A .

Find the value of x .

(6)



When a force of 150 N is applied vertically upwards at B , the branch is on the point of tilting about C .

When a force of 225 N is applied vertically downwards at B , the branch is on the point of tilting about D .

The branch is modelled as a non-uniform rod AB of weight W newtons.

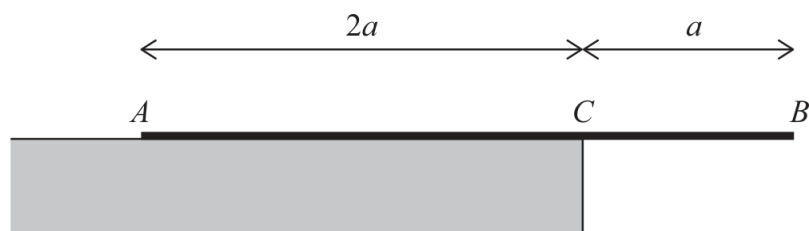
The distance from the point C to the centre of mass of the rod is x metres.

Use the model to find

- (i) the value of W
- (ii) the value of x

(8)

5.



A non-uniform rod AB of length $3a$ rests in equilibrium on a horizontal ledge and overhangs the edge of the ledge at C .

The point C is such that $AC = 2a$ and $CB = a$, as shown in Figure 3.

The rod has weight W .

The distance of the centre of mass of the rod from A is d .

The rod is perpendicular to the edge of the ledge.

When a force of magnitude P , **acting vertically upwards**, is applied to the rod at B , the rod is on the point of tilting about A .

When the force applied at B is replaced by a force of magnitude $1.25P$, **acting vertically downwards** at B , the rod is on the point of tilting about C .

Find d in terms of a .

(6)